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SIGMA-ALDRICH®

Functional Genomics & RNAi

Preparation of the Lentiviral Transduction Particles Using Packaging Plasmid Mix

Product No. **SHP001**

Protocol provided by MISSION® Team

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Introduction

The MISSION® Lentiviral Packaging Mix is an optimized formulation of two plasmids expressing the key HIV packaging genes and a heterologous viral envelope gene. MISSION® lentiviral particles are generated from three main components:

1. The packaging vector, which contains the minimal set of lentiviral genes required to generate the virion structural proteins and packaging functions.
2. The vesicular stomatitis virus G-protein (pCMV-VSVG) envelope vector, which provides the heterologous envelope for pseudo-typing.
3. The shRNA transfer vector (pLK0.1), which contains the sequence of interest as well as the cis acting sequences necessary for RNA production and packaging.

Materials

- 293T cells (in log growth and at 70% confluence on the day of transfection)
- Media - DME media ([D6429](#)) supplemented with 10% fetal bovine serum ([F2442](#)) and 4 mM L-glutamine ([G6392](#))
- Transfection Reagent
- 96 well tissue culture plates
- Tissue culture incubator—37 °C, 5% CO₂, 100% relative humidity
- Tissue culture laminar flow hood—only open optimization plate in tissue culture hood
- Lentiviral Packaging Mix ([SHP001](#))

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Protocol

Day 0

- 0.1 Seed HEK293T cells (20,000 cells/well of 96-well plate) in complete DME media.

Day 1

- 1.1 Thaw the vial of Lentiviral Packaging Mix at room temperature and place on ice.
- 1.2 Put three empty polypropylene tubes on ice.
- 1.3 Add 1 μ l of Lentiviral Packaging Mix per well to the first tube.
- 1.4 Add 15 μ l of serum free DME and 0.6 μ l transfection reagent, Fugene 6, per well into the second tube.
- 1.5 Add 0.1 μ g/well of shRNA transfer vector into the third tube.
- 1.6 Combine all transfection cocktail components together in one polypropylene vial.
- 1.7 Mix gently by pipetting up and down.
- 1.8 Incubate at room temperature for 15 min.
- 1.9 Equally divide the mixture among the well to be transfected.
- 1.10 Incubate cells at 37 °C overnight in tissue culture incubator.

Day 2

- 2.1 16 hours post-transfection, gently remove media from the transfected cells (avoid disturbing cells) and replace with 100 μ l of the pre-warmed complete media per well of 96-well plate.
- 2.2 Incubate cells in incubator (37 °C and 5% CO₂) for an additional 24 hours.

Day 3

- 3.1 Double glove before proceeding to 3.2.
- 3.2 Gently remove supernatant, now containing virus, and place it in a collection plate.
- 3.3 Add fresh complete DME to the wells and incubate cells at 37 °C overnight.
- 3.4 Cover, and store collection plate at 4 °C.

Day 4

- 4.1 Double glove before proceeding to 4.2.
- 4.2 Remove second viral supernatant from wells and add it to the collection plate from the day before.
- 4.3 Determine titer by p24 ELISA.
- 4.4 Store virus tightly sealed at –80 °C.

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Tips

It was determined experimentally that the first harvest's titer is slightly higher than the second, but pooling the two will not significantly decrease the overall titer of the sample.

Day 0

- HEK293T cells should be maintained for a limited number of passages.
- Seed cells overnight, no longer than 24 hours before transfection.

Day 1

- Check the volume and appearance of the product.
- Vector has to be prepared using Endo-free purification procedure for the best titer and reproducibility results. Use only sterile polypropylene tube(s) to prevent contamination.

- Intensive mixing can damage formation of the transfection complex. Do not disturb the tubes during incubation because the transfection complex is formed during that time.
- Add the transfection cocktail to the cells gently, distributing drops evenly in a clockwise motion from the center to the sides of the wells.

Day 2

- The first viral collection can be stored at 2–8 °C for 24–48 hours.

Day 4

- Each freeze-thaw cycle can cause subsequent titer reduction. It is recommended to prepare small aliquots of viral stock and store at –70 °C until ready for use.

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